

Assignment 1: Due April 14th, 2026

Turn-in

Submit a PDF with answers to each of the following questions, include your name, NetID, and PERM number in the submission. Submissions will be taken via Gradescope, instructions on how to join the course on Gradescope and submit the assignment are posted below. Please keep your answers as precise and concise as possible. Late submissions will be acceptance and deducted in accordance to the course late-day policy.

Overview

In this assignment, we'll investigate the 802.11 wireless network protocol. This assignment is meant to serve as a hands-on introduction to wireless protocol, before we cover the history and design decisions in lecture. A couple useful resources to refer to during this assignment are [A Technical Tutorial on the 802.11 Protocol](#), by Pablo Brenner (Breezecom Communications), and of course, there is the 802.11 standard itself, [ANSI/IEEE Std 802.11, 1999 Edition \(R2003\)](#). In particular, you may find Table 1 on page 36 of the standard particularly useful when looking through the wireless trace. We will provide a trace of captured 802.11 frames for you to analyze which you are to use to answer the following questions.

Getting Started

Download [this trace file](#) `Wireshark_802_11.pcap`. It was collected using AirPcap and Wireshark running on a computer in a home network. The network consisted of a Linksys 802.11g combined access point/router, with two wired PCs and one wireless host PC attached to the access point/router. There were access points in neighboring houses available as well, and can be overheard in the trace. In this trace file, we'll see frames captured on channel 6. Since the host and AP that we are interested in are not the only devices using channel 6, we'll see a lot of frames that we're not interested in for this lab, such as beacon frames advertised by a neighbor's AP also operating on channel 6. The wireless host activities taken in the trace file are:

- The host is already associated with the 30 Munroe St AP when the trace begins.
- At $t = 24.81$, the host makes an HTTP request to `http://gaia.cs.umass.edu/wiresharklabs/alice.txt`. The IP address of `gaia.cs.umass.edu` is `128.119.245.12`.
- At $t=32.82$, the host makes an HTTP request to `http://www.cs.umass.edu`, whose IP address is `128.119.240.19`.
- At $t = 49.58$, the host disconnects from the 30 Munroe St AP and attempts to connect to `linksys_ses_24086`. This is not an open access point and so the host is eventually unable to connect to this AP.
- At $t=63.14$ the host gives up trying to associate with the `linksys_ses_24086` AP, and associates again with the 30 Munroe St access point.

Once you have downloaded the trace, you can load it into Wireshark and view the trace using the File pull down menu, choosing Open, and then selecting the `Wireshark_802_11.pcap` trace file.

Beacon Frames

Recall that beacon frames are used by an 802.11 AP to advertise its existence. To answer some of the questions below, you'll want to look at the details of the IEEE 802.11 frame and subfields in the middle Wireshark window.

1. What is the SSID of the access point that is issuing most of the beacon frames in this trace?
2. What are the intervals of time between the transmission of the beacon frames from the 30 Munroe St. access point? (Hint: this interval of time is contained in the beacon frame itself).
3. What (in hexadecimal notation) is the source MAC address on the beacon frame from 30 Munroe St? Recall that the source, destination, and BSS are three addresses used in an 802.11 frame. For a detailed discussion of the 802.11 frame structure, see section 7 in the IEEE 802.11 standards document (cited above).
4. What (in hexadecimal notation) is the destination MAC address on the beacon frame from 30 Munroe St?
5. The beacon frames from the 30 Munroe St access point advertise that the access point can support four data rates and eight additional *extended supported rates*. What are these rates?

Data Transfer

Since the trace starts with the host already associated with the AP, let's first look at data transfer over an 802.11 association before looking at AP association/disassociation. Recall that in this trace, at $t = 24.82$, the host makes an HTTP request to `http://gaia.cs.umass.edu/wiresharklabs/alice.txt`. The IP address of `gaia.cs.umass.edu` is 128.119.245.12. Then, at $t=32.82$, the host makes an HTTP request to `http://www.cs.umass.edu`. Make sure that your Time Display Format is *Seconds Since Beginning of Capture* to make it easier to find these packets.

1. Find the 802.11 frame containing the SYN TCP segment for this first TCP session (that downloads `alice.txt`).
2. What are three MAC address fields in the 802.11 frame?
3. In this frame, write the MAC address (in hexadecimal representation) corresponding to the wireless host, the access point and the first-hop router.
4. What is the IP address of the wireless host sending this TCP segment?
5. What is the destination IP address?
6. Does this destination IP address correspond to the host, access point, first-hop router, or some other network-attached device? Explain.

Association/Disassociation

Recall that a host must first associate with an access point before sending data. Association in 802.11 is performed using the ASSOCIATE REQUEST frame (sent from host to AP, with a frame type 0 and subtype 0) and the ASSOCIATE RESPONSE frame (sent by the AP to a host with a frame type 0 and subtype of 1, in response to a received ASSOCIATE REQUEST). For a detailed explanation of each field in the 802.11 frame, see page 34 (Section 7) of the 802.11 spec at <http://gaia.cs.umass.edu/wireshark-labs/802.11-1999.pdf>.

1. What two actions are taken (i.e., frames are sent) by the host in the trace just after $t=49.5$, to end the association with the 30 Munroe St AP that was initially in place

when trace collection began? (Hint: one is an IP-layer action, and one is an 802.11-layer action).

2. An ASSOCIATE REQUEST from host to AP, and a corresponding ASSOCIATE RESPONSE frame from AP to host are used for the host to associated with an AP. At what time is there an ASSOCIATE REQUEST from host to the 30 Munroe St AP? When is the corresponding ASSOCIATE REPLY sent? (Note that you can use the filter expression `wlan.fc.subtype < 2 and wlan.fc.type == 0 and wlan.addr == IntelCor_d1:b6:4f` to display only the ASSOCIATE REQUEST and ASSOCIATE RESPONSE frames for this trace.)
3. What transmission rates is the host willing to use? The AP? To answer this question, you will need to look into the parameters fields of the 802.11 wireless LAN management frame.

Other Frames

This trace contains a number of PROBE REQUEST and PROBE RESPONSE frames.

1. What are the sender, receiver and BSS ID MAC addresses in these frames?
2. What is the purpose of these two types of frames? (To answer this last question, you'll need to dig into the online references cited earlier in this lab).

Gradescope

Submissions will made done through Gradescope as a PDF. You can enroll in the course through Gradescope using the entry code **4DWRD5**. Please use either your @ucsb.edu or @umail.ucsb.edu email when joining the course.